# Lab Five-Six – Data Wrangling in R

**Tips**

- Complete the tasks below. Make sure to start your solutions in on a new line that starts with "**Solution**:".
- Make sure to use the Quarto Cheatsheet. This will make completing and writing up the lab *much* easier.
- Make sure to use "enter" to make your code readable, so it doesn't run off the page. I switched the Quarto document to automatically render to .pdf, so you can see what you're submitting by default. You can switch back for the highlighting by moving the html block above the pdf block in the YAML header.
- Use the pipe operator (`|>`). It makes the data wrangling tasks much easier to write.
- Use `head(...)` or `glimpse(...)` often to ensure your code did what you wanted it to and that you haven't accidentally deleted their entire dataset with a bad filter.
- Don't forget that you can copy-and-paste code when you're doing something similar as we did in class!

**Timeline**

- Week of 2/16: Aim to complete the data wrangling sections
- Week of 2/23: Aim to complete the data summaries sections

# 1 Biomedical Science Data

Kitsberg et al. (2025) collect data on the Human Cytomegalovirus (HCMV). HCMV is a member of the herpesvirus family – the same family that includes: chickenpox, mononucleosis (mono), and cold sores. Prevalence is high with a worldwide seroprevalence of 24% to 100% (Gupta and Shorman 2025). The virus persists in a dormant state for the entire life of those infected. Primary cytomegalovirus infection is often asymptomatic but can present with flu-like symptoms. The infection can reactivate due to stress, aging, or a weakened immune system.

The researchers obtained blood samples from healthy donors, males and females, aged 25-45. They infected the cells of primary (human donor's blood) or THP1 (purchased cell lines) monocytes ($n = 87$ and $n = 112$, respectively) and macrophages ($n = 93$ and $n = 170$, respectively) and counted the number of viral genomes in the nuclei.

Note that the original donor of the THP-1 cell line was a 1-year-old Japanese boy. He was treated for Acute Monocytic Leukemia at Tohoku Hospital Pediatrics (THP-1; the first cell line there). Because they are malignant cancer cells, they can divide indefinitely, breaking the usual cell managment that usually reigns in rogue cells. Therefore, these cells can divide indefinitely in a lab environment. THP-1 cells, and those like them, serve as such a reliable, uniform model for medical research and improve reproducability.

**Preliminary Question**: Are THP-1 cells a good stand-in for human cells?

**Research Question**: Is the virus more prevalent in the nucleus of monocytes or macrophages, explaining why HCMV stays dormant in monocytes but goes active in macrophages?

## 1.1 Data Wrangling

### 1.1.1 Part A

Go to the website and download the source data https://www.nature.com/articles/s41467-025-68063-y#Sec30.

Describe the process – do not leave out any steps – for how you can load the data for Figures 3b and 3d from the manuscript into `R`.

**Solution**

First, locate the source data within the nature article linked at `Source Data`. From the provided excel document, save the excel sheet `Fig. 3B` to a separate excel document. Within this document, first delete the empty top row, and the Primary and THP1 cells. Next, cut and paste the entire data contents of the two right columns below the last entry on the left columns. After this, the data should be loaded into a csv file with two columns, I renamed the column headers to be `sample` and `count` respectively.

**Hint**: You can use packages like `readxl` (Wickham and Bryan 2025) but you will find it substantially easier to open the `.xls` file using your computer's spreadsheet software and create a new `.csv` file with the data you need.

### 1.1.2 Part B

The data need to be cleaned. Note that the sample type (Healthy Donor or THP-1), cell type (Monocytes or Macrophages), and treatment (12 Hours Post-Infection or Uninfected) are all recorded in one column `sample`, separated with underscores. Use `separate(...)` from the `tidyverse` package to separate the `sample` column into the `SampleType`, `CellType`, `Treatment` columns. Save the object as `dat.kitsberg.cleaned`.

**Solution**

```
dat.kitsberg = read_csv("fig3b.csv")
dat.kitsberg.cleaned = dat.kitsberg |>
  separate_wider_delim("sample","_",names=c("SampleType","CellType","Treatment"))
head(dat.kitsberg.cleaned)
```

```
# A tibble: 6 x 4
  SampleType CellType Treatment  count
  <chr>      <chr>    <chr>      <dbl>
1 primary    mono     uninfected     0
2 primary    mono     uninfected     0
3 primary    mono     uninfected     0
4 primary    mono     uninfected     0
5 primary    mono     uninfected     0
6 primary    mono     uninfected     0
```

**Hint**: Use `?separate(...)` to see how it works.

### 1.1.3 Part C

The researchers are only interested in observations in the treatment group (12hpi). Remove uninfected samples from the data. Update `dat.kitsberg.cleaned` with the result.

**Solution**

```
dat.kitsberg.cleaned = dat.kitsberg.cleaned |>
  filter(Treatment != "uninfected")

head(dat.kitsberg.cleaned)
```

```
# A tibble: 6 x 4
  SampleType CellType Treatment count
  <chr>      <chr>    <chr>     <dbl>
1 primary    mono     12hpi         0
2 primary    mono     12hpi         0
3 primary    mono     12hpi         0
4 primary    mono     12hpi         0
5 primary    mono     12hpi         0
6 primary    mono     12hpi         0
```

**Hint**: I would use the `filter(...)` function.

### 1.1.4 Part D

When `CellType` is "MDM", it is a Monocyte-Derived Macrophages. That is, the cell type should be macrophages which is coded as "mac". Update `dat.kitsberg.cleaned` with the result.

**Solution**

```
dat.kitsberg.cleaned = dat.kitsberg.cleaned |>
  mutate(CellType = case_when(
                  (CellType == 'MDM' | CellType == 'mac') ~ 'mac',
                  CellType == 'mono' ~ 'mono',
                  CellType == 'PDGFR' ~ 'PDGFR'))

unique(dat.kitsberg.cleaned$CellType)
```

```
[1] "mono"  "mac"   "PDGFR"
```

**Hint**: I would use the `mutate(...)` function. You will likely find `if_else(...)` or `case_when(...)` helpful.

## 1.2 Part E

The researchers are only interested in monocytes and macrophages. Remove any other cell types. Update `dat.kitsberg.cleaned` with the result.

**Solution**

```
dat.kitsberg.cleaned = dat.kitsberg.cleaned |>
  filter(CellType %in% c("mono","mac"))

unique(dat.kitsberg.cleaned$CellType)
```

```
[1] "mono" "mac"
```

**Hint**: I would use the `filter(...)` function.

## 1.3 Part F

Clean up the data entry. Replace the shorthand the researchers used with the neat sample type (Healthy Donor or THP-1), cell type (Monocytes or Macrophages), and treatment (12 Hours Post-Infection or Uninfected). Update `dat.kitsberg.cleaned` with the result.

**Solution**

```
dat.kitsberg.cleaned = dat.kitsberg.cleaned |>
  mutate(SampleType = case_when(
        SampleType == "primary" ~ "Healthy Donor",
        SampleType == "THP1" ~ "THP-1")) |>
  mutate(CellType = case_when(
        CellType == "mono" ~ "Monocytes",
        CellType == "mac" ~ "Macrophages")) |>
  mutate(Treatment = "12 Hours Post-Infection") #all uninfected already removed

head(dat.kitsberg.cleaned)
```

```
# A tibble: 6 x 4
  SampleType    CellType  Treatment               count
  <chr>         <chr>     <chr>                   <dbl>
1 Healthy Donor Monocytes 12 Hours Post-Infection     0
2 Healthy Donor Monocytes 12 Hours Post-Infection     0
3 Healthy Donor Monocytes 12 Hours Post-Infection     0
4 Healthy Donor Monocytes 12 Hours Post-Infection     0
5 Healthy Donor Monocytes 12 Hours Post-Infection     0
```

```
6 Healthy Donor Monocytes 12 Hours Post-Infection     0
```

**Hint**: I would use the `mutate(...)` function. You will likely find `if_else(...)` or `case_when(...)` helpful.

## 1.4 Part G

In an `R` code block, combine your answers into one code block that completes the full cleaning process. Save the object as `dat.kitsberg.cleaned`.

**Solution**

```
dat.kitsberg.cleaned = dat.kitsberg |>
  separate_wider_delim("sample","_",names=c("SampleType","CellType","Treatment")) |>
  filter(Treatment != "uninfected") |> #remove uninfected
  mutate(CellType = case_when(
        (CellType == 'MDM' | CellType == 'mac') ~ 'mac',
        CellType == 'mono' ~ 'mono')) |> #rename MDM to mac
  filter(CellType %in% c("mono","mac")) |> #only keep mono and mac
  #prettier data entry -------------
  mutate(SampleType = case_when(
        SampleType == "primary" ~ "Healthy Donor",
        SampleType == "THP1" ~ "THP-1")) |>
  mutate(CellType = case_when(
    CellType == "mono" ~ "Monocytes",
    CellType == "mac" ~ "Macrophages")) |>
  mutate(Treatment = "12 Hours Post-Infection") #all uninfected already removed
head(dat.kitsberg.cleaned)
```

```
# A tibble: 6 x 4
  SampleType    CellType  Treatment                count
  <chr>         <chr>     <chr>                    <dbl>
1 Healthy Donor Monocytes 12 Hours Post-Infection     0
2 Healthy Donor Monocytes 12 Hours Post-Infection     0
3 Healthy Donor Monocytes 12 Hours Post-Infection     0
4 Healthy Donor Monocytes 12 Hours Post-Infection     0
5 Healthy Donor Monocytes 12 Hours Post-Infection     0
6 Healthy Donor Monocytes 12 Hours Post-Infection     0
```

**Note**: You will use this in the next section

## 1.5 Data Summaries

### 1.5.1 Part A

Recreate Figure 3b as depicted in the paper (see https://www.nature.com/articles/s41467-025-68063-y#Fig3). You can ignore the significance comparison because we haven't done that yet.

**Solution**

The paper's original plot included a jitter overlaying the violin, and although this further emphasizes the cluster around 0, I felt that the information provided didn't justify the added noise of the graph.

**Hint**: I would use `geom_violin(...)` for Figure 3b I also expect your plots to look nicer – better labels, better scaling, etc. Don't forget to use the proper formatting using the syntax in the Quarto cheatsheet.

### 1.5.2 Part B

Compute a table containing relevant numerical summaries for Figure 3b.

**Solution**

Figure 1: HCMV Genomes in Nucleus vs Sample Type

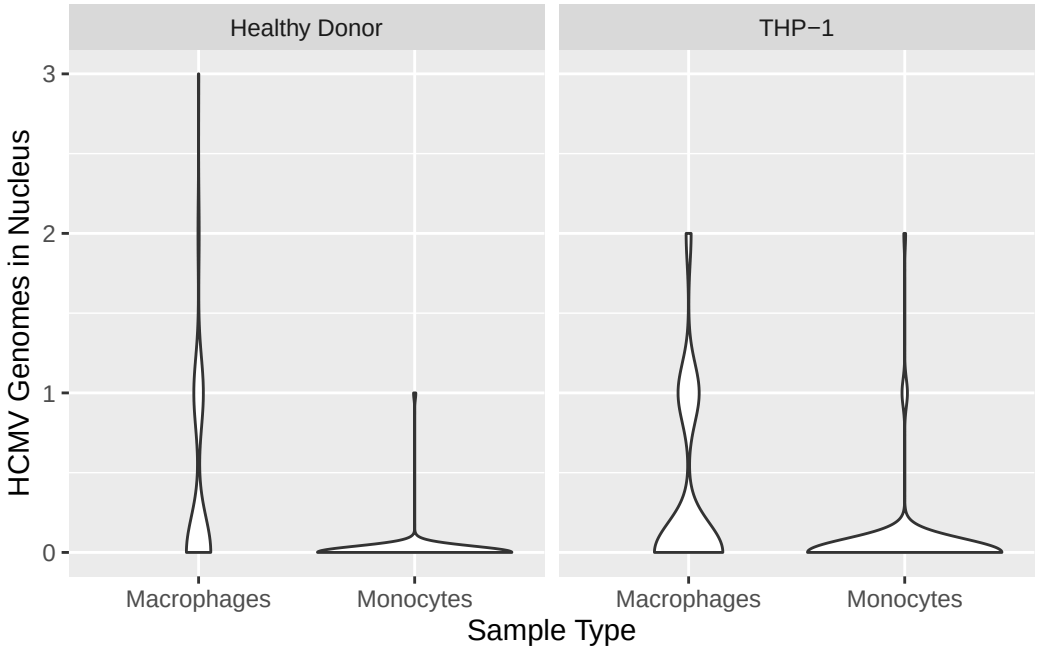



Table 1

| SampleType | CellType | n | Mean | SD | Median | IQR |
|---|---|---|---|---|---|---|
| Healthy Donor | Macrophages | 93 | 0.37 | 0.60 | 0 | 1 |
| Healthy Donor | Monocytes | 87 | 0.01 | 0.11 | 0 | 0 |
| THP-1 | Macrophages | 170 | 0.33 | 0.57 | 0 | 1 |
| THP-1 | Monocytes | 112 | 0.04 | 0.25 | 0 | 0 |

As shown in Table 1, the count of both cell types across both samples is crowded around 0, while the means are higher for Macrophages across both cell types. The means differ by .03 across sample types for both cell types, which is quite small given that the variable of interest is a count variable.

**Hint**: I would use `group_by(...)` and `summarize(...)` to complete this. Don't forget to use the proper formatting using `knitr::kable(...)` as described in the Quarto Cheatsheet.

# 2 Ship Valuation

Figure 2: This is a ship-shipping ship shipping shipping ships.



Bal et al. (2025) aim to build a model for predicting the value of Supramax and Ultramax ships using both current and past market data. Their goal was to build a model that only uses information anyone can easily look up.

The researchers collected 17 years of real ship sale prices (2005–2022), covering $n = 1819$ Supramax and Ultramax ship sales.

**Research Question**: Can we predict Supramax and Ultramax ship prices?

## 2.1 Data Wrangling

### 2.1.1 Part A

I downloaded the data from linked in the article (https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0319073). I have included the data in this lab folder as `ShipValuation-Bal25.csv`. Load the data into `R`.

### 2.1.2 Part B

The price of each Supramax and Ultramax ship is reported on the $\log_{10}$ scale. Create a new column called `price` that contains the price in millions of dollars.

$$\text{Price} = \frac{10^{\text{SPLog}}}{1000000}$$

Save the object as `dat.bal.cleaned`.

**Solution**

```
dat.bal = read_csv("ShipValuation-Bal25.csv")[1:1819, ]
dat.bal.cleaned = dat.bal |>
  mutate(price = (10^SPLog) / 1000000)

head(dat.bal.cleaned$price)
```

```
[1]  8.20000  5.50000 15.00000  7.20512  5.00000  7.00000
```

200 rows were removed from the dataset due to having missing values.

**Hint**: I would use the `mutate(...)` function.

### 2.1.3 Part C

The authors recorded the country that each Supramax or Ultramax ship was built in in the columns `Japan`, `China, People's Republic Of`, `Philippines`, `Korea, South`, `Others Country Build`. These columns are marked `1` to indicate that is the country of origin and `0` otherwise. Create a new column called `OriginCountry` that contains a clean labels for the country in which the ship was built. Update `dat.bal.cleaned` with the result.

**Solution**

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(OriginCountry = case_when(
    Japan == 1 ~ "Japan",
    `China, People's Republic Of` == 1 ~ "China, People's Republic Of",
    Philippines == 1 ~ "Philippines",
    `Korea, South` == 1 ~ "Korea, South",
    TRUE ~ "Others Country Build"))
unique(dat.bal.cleaned$OriginCountry)
```

```
[1] "Others Country Build"        "Japan"
[3] "Korea, South"                "China, People's Republic Of"
[5] "Philippines"
```

**Hint**: I would use the `mutate(...)` function. You will likely find `if_else(...)` or `case_when(...)` helpful.

### 2.1.4 Part D

The authors recorded each Supramax and Ultramax ship's main engine manufacturer in the columns `Caterpillar`, `MAN-B&W`, `Wartsila`, `Mitsubishi`. These columns are marked `1` to indicate that is the manufacturer and `0` otherwise. Create a new column called `Manufacturer` that contains a clean labels for the manufacturer. Update `dat.bal.cleaned` with the result.

**Solution**

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(Manufacturer = case_when(
        Caterpillar == 1 ~ "Caterpillar",
        `MAN-B&W` == 1 ~ "Man-B&W",
        Wartsila == 1 ~ 'Wartsila',
        Mitsubishi == 1 ~ 'Mitsubishi'))
unique(dat.bal.cleaned$Manufacturer)
```

```
[1] "Man-B&W"     "Wartsila"    "Mitsubishi"  "Caterpillar"
```

**Hint**: Follow the logic of the previous question.

### 2.1.5 Part E

The researchers recorded whether each Supramax and Ultramax ship is currently covered by a Protection and Indemnity (P&I) Club. This can be an indicator that a ship meets the minimum seaworthiness and safety requirements of the insurer.

The authors recorded P&I clubs in the columns `Japan P&I Club`, `Gard`, `West of England`, `North of England`, `UK P&I`, `Britannia P&I`, `London P&I Club`, `Skuld`, `Steamship Mutual`, `Others`, `Unknown`. These columns are marked `1` to indicate that the ship is insured by that P&I club and `0` otherwise. Create a new column called `PI.club` that contains a clean labels for the P&I club. Update `dat.bal.cleaned` with the result.

**Solution**

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(PI.club = case_when(
    `Japan P&I Club` == 1 ~ "Japan P&I Club",
    Gard == 1 ~ "Gard",
    `North of England` == 1 ~ 'North of England',
```

```
    `West of England` == 1 ~ "West of England",
    `UK P&I` == 1 ~ "UK P&I",
    `Britannia P&I` == 1 ~ "Britannia P&I",
    `London P&I Club` == 1 ~ "London P&I Club",
     Skuld == 1 ~ "Skuld",
     `Steamship Mutual` == 1 ~ "Steamship Mutual",
     Others == 1 ~ "Others",
     Unknown == 1 ~ "Unknown"))
unique(dat.bal.cleaned$PI.club)
```

```
 [1] "Steamship Mutual" "West of England"  "UK P&I"           "Others"
 [5] "Gard"             "North of England" "Britannia P&I"    "London P&I Club"
 [9] "Unknown"          "Japan P&I Club"   "Skuld"
```

**Hint**: Follow the logic of the previous questions.

### 2.1.6 Part F

The researchers recorded whether each Supramax and Ultramax ship is classed by a society that requires a ship to be safe, structurally sound, and built/maintained to specific engineering standards.

The authors recorded class society in the columns `Nippon Kaiji Kyokai (IACS)`, `Lloyd's Register`, `Bureau Veritas (IACS)`, `Det Norske Veritas (IACS)`, `American Bureau of Shipping (IACS)`, `Others Class`, `With Two Classes`, `Disclassed`. These columns are marked `1` to indicate that the ship is classed by that soceity and `0` otherwise. Create a new column called `Class` that contains a clean labels for the class. Update `dat.bal.cleaned` with the result.

**Solution**

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(Class = case_when(
    `Nippon Kaiji Kyokai (IACS)` == 1 ~ "Nippon Kaiji Kyokai (IACS)",
    `Lloyd's Register` == 1 ~ "Lloyd's Register",
    `Bureau Veritas (IACS)` == 1 ~ "Bureau Vertias (IACS)",
    `Det Norske Veritas (IACS)` == 1 ~ "Det Norske Veritas (IACS)",
    `American Bureau of Shipping (IACS)` == 1 ~ "American Bureau of Shipping (IACS)",
    `Others Class` == 1 ~ "Others Class",
    `With Two Classes` == 1 ~ "With Two Classes",
    Disclassed == 1 ~ "Disclassed"))

unique(dat.bal.cleaned$Class)
```

```
[1] "Others Class"                       "Disclassed"
[3] "American Bureau of Shipping (IACS)" "Bureau Vertias (IACS)"
[5] "Det Norske Veritas (IACS)"          "Lloyd's Register"
[7] "Nippon Kaiji Kyokai (IACS)"         "With Two Classes"
```

**Hint**: Follow the logic of the previous questions.

### 2.1.7 Part G

The researchers recorded the flag state of each Supramax and Ultramax ship. Each flag state enforces its own set of safety, environmental, and crew welfare rules and regulations. Ships under certain flag states may be more or less valuable depending on how much the flag affects the buyer's trust in the condition of the boat.

The authors recorded flag states in the columns `Panama`, `Marshall Islands`, `Liberia`, `Malta`, `Hong Kong`, `China`, `Singapore`, `Others Flag`, `With two flags`. These columns are marked `1` to indicate that the ship has that flag state and `0` otherwise. Create a new column called `FlagState` that contains a clean labels for the flag state. Update `dat.bal.cleaned` with the result.

**Note**: There is a minor issue in the above text, the corect flag state is `Hong Kong, China`

**Solution**

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(FlagState = case_when(
      Panama == 1 ~ "Panama",
      `Marshall Islands` == 1 ~ "Marshall Islands",
      Liberia == 1 ~ "Liberia",
      Malta == 1 ~ "Malta",
      `Hong Kong, China` == 1 ~ "Hong Kong, China",
      Singapore == 1 ~ "Singapore",
      `Others Flag` == 1 ~ "Others Flag",
      `With two flags` == 1 ~ "With two flags"))

unique(dat.bal.cleaned$FlagState)
```

```
[1] "Panama"           "Others Flag"      "Malta"            "Liberia"
[5] "Hong Kong, China" "Marshall Islands" "Singapore"        "With two flags"
```

**Hint**: Follow the logic of the previous questions.

## 2.2 Part H

In an `R` code block, combine your answers into one code block that completes the full cleaning process. Save the object as `dat.bal.cleaned`.

**Solution**

```
dat.bal.cleaned = dat.bal |>
  mutate(price = (10^SPLog) / 1000000) |>

  mutate(OriginCountry = case_when(
    Japan == 1 ~ "Japan",
    `China, People's Republic Of` == 1 ~ "China, People's Republic Of",
    Philippines == 1 ~ "Philippines",
    `Korea, South` == 1 ~ "Korea, South",
    TRUE ~ "Others Country Build")) |>

  mutate(Manufacturer = case_when(
    Caterpillar == 1 ~ "Caterpillar",
    `MAN-B&W` == 1 ~ "Man-B&W",
    Wartsila == 1 ~ 'Wartsila',
    Mitsubishi == 1 ~ 'Mitsubishi')) |>

  mutate(PI.club = case_when(
    `Japan P&I Club` == 1 ~ "Japan P&I Club",
    Gard == 1 ~ "Gard",
    `North of England` == 1 ~ 'North of England',
    `West of England` == 1 ~ "West of England",
    `UK P&I` == 1 ~ "UK P&I",
    `Britannia P&I` == 1 ~ "Britannia P&I",
    `London P&I Club` == 1 ~ "London P&I Club",
    Skuld == 1 ~ "Skuld",
    `Steamship Mutual` == 1 ~ "Steamship Mutual",
    Others == 1 ~ "Others",
    Unknown == 1 ~ "Unknown")) |>

  mutate(Class = case_when(
    `Nippon Kaiji Kyokai (IACS)` == 1 ~ "Nippon Kaiji Kyokai (IACS)",
    `Lloyd's Register` == 1 ~ "Lloyd's Register",
    `Bureau Veritas (IACS)` == 1 ~ "Bureau Vertias (IACS)",
```

```
    `Det Norske Veritas (IACS)` == 1 ~ "Det Norske Veritas (IACS)",
    `American Bureau of Shipping (IACS)` == 1 ~ "American Bureau of Shipping (IACS)",
    `Others Class` == 1 ~ "Others Class",
    `With Two Classes` == 1 ~ "With Two Classes",
    Disclassed == 1 ~ "Disclassed")) |>

  mutate(FlagState = case_when(Panama == 1 ~ "Panama",
  `Marshall Islands` == 1 ~ "Marshall Islands",
  Liberia == 1 ~ "Liberia",
  Malta == 1 ~ "Malta",
  `Hong Kong, China` == 1 ~ "Hong Kong, China",
  Singapore == 1 ~ "Singapore",
  `Others Flag` == 1 ~ "Others Flag",
  `With two flags` == 1 ~ "With two flags"))
```

**Note**: You will use this in the next section

## 2.3 Data Summaries

First, I provide a short description of the other variables in their data. Note that the size ordering of types of ships discussed below is:

$$\text{Handysize} < \text{Supramax} < \text{Ultramax} < \text{Panamax} < \text{Capesize}$$

- `DWT`: The deadweight tonnage, which is the maximum weight each Supramax and Ultramax ship can safely carry in metric tons. This gives a measure of how much cargo, fuel, fresh water, ballast water, provisions, and crew a ship can handle.
- `Yas`: The age in years. New Supramax and Ultramax ships were recorded as having age 1.
- `Enginges RPM`: Revolutions per minute of each Supramax and Ultramax ship's main propulsion engine. Generally speaking, lower-RPM engines are more expensive because they are more fuel efficient.
- `BACI`: The Baltic Capesize Index. This describes freight rates for Capesize dry bulk ships, which reflects supply and demand. When BACI is high, Capesize ships are more profitable in shipping, which could positively affect Supramax and Ultramax ship values as demand for dry bulk commodities might be high and large shipments could be split into several Supramax and Ultramax shipments.
  **Note**: This reports the current BACI. The data also contain the BACI form 1, 2, and 3 months before the sale (`BACI-1`, `BACI-2`, `BACI-3`).
- `BPNI`: The Baltic Panamax Index. This describes the cost to ship dry bulk commodities (e.g., coal or grain) on Panamax class ships. When BPNI is high, we expect Panamax ship prices to go up. Supramax and Ultramax ship values tend to follow this trend as charterers switch to these smaller vessels when Panamax rates become too expensive.
  **Note**: This reports the current BPNI. The data also contain the BPNI form 1, 2, and 3 months before the sale (`BPNI-1`, `BPNI-2`, `BPNI-3`).
- `BSIS`: The Baltic Supramax Index. This describes freight rates for Supramax vessels, which reflects supply and demand. When BSIS is high, Supramax and Ultramax prices are usually higher because they are more profitable in shipping.
  **Note**: This reports the current BSIS. The data also contain the BSIS form 1, 2, and 3 months before the sale (`BSIS-1`, `BSIS-2`, `BSIS-3`).
- `BHSI`: The Baltic Handysize Index. This describes freight rates for handysize vessels. When BHSI is high, Supramax and Ultramax ship prices are usually higher because the prices move together.
  **Note**: This reports the current BHSI. The data also contain the BHSI form 1, 2, and 3 months before the sale (`BHSI-1`, `BHSI-2`, `BHSI-3`).
- `LIBOR`: The London InterBank Offered Rate. This describes the average reported interest rate at which banks would lend US dollars for 3 months.

### 2.3.1 Part A

Choose a categorical variable. Create a plot and corresponding numerical summary to assess whether or how ship prices vary across the levels of the selected categorical variable.

**Solution**

Figure 3: Ship Price vs P&I Club Membership

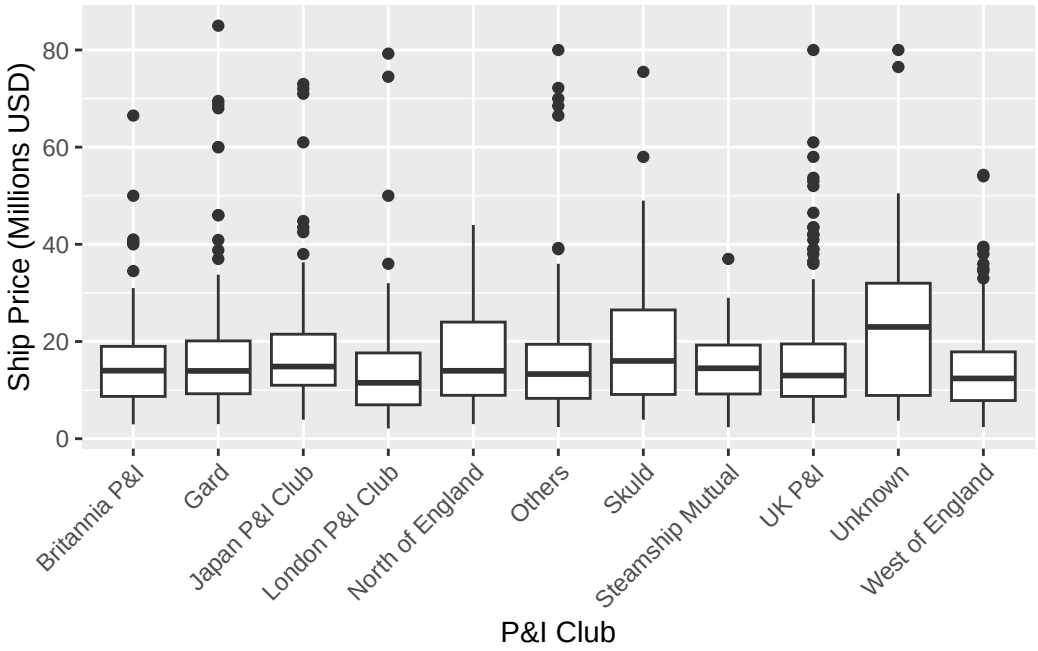



Table 2: Ship Price (Millions USD) by P&I Club

| P&I Club | Mean | SD | Median | IQR |
|---|---|---|---|---|
| Britannia P&I | 15.51 | 9.45 | 14.00 | 10.30 |
| Gard | 16.68 | 11.84 | 13.95 | 10.88 |
| Japan P&I Club | 17.17 | 9.99 | 14.85 | 10.50 |
| London P&I Club | 14.31 | 11.37 | 11.50 | 10.67 |
| North of England | 17.23 | 10.40 | 13.97 | 15.08 |
| Others | 15.73 | 11.01 | 13.30 | 11.13 |
| Skuld | 19.22 | 12.41 | 16.00 | 17.40 |
| Steamship Mutual | 14.86 | 7.07 | 14.50 | 10.07 |
| UK P&I | 16.95 | 12.70 | 13.00 | 10.80 |
| Unknown | 22.99 | 15.05 | 23.00 | 23.10 |
| West of England | 14.18 | 8.79 | 12.40 | 10.02 |

As seen in both Figure 3 and Table 2, P&I clubs insure ships of a similar distribution of prices, suggesting that insurers tend to not only diversify their portfolios, but also have essentially reached a consensus on the appropriate basket of ships.

### 2.3.2 Part B

Choose a quantitative variable. Create a plot and corresponding numerical summary to assess whether or how ship prices vary with the selected quantitative variable.

**Solution**

Figure 4: Ship Price vs LIBOR

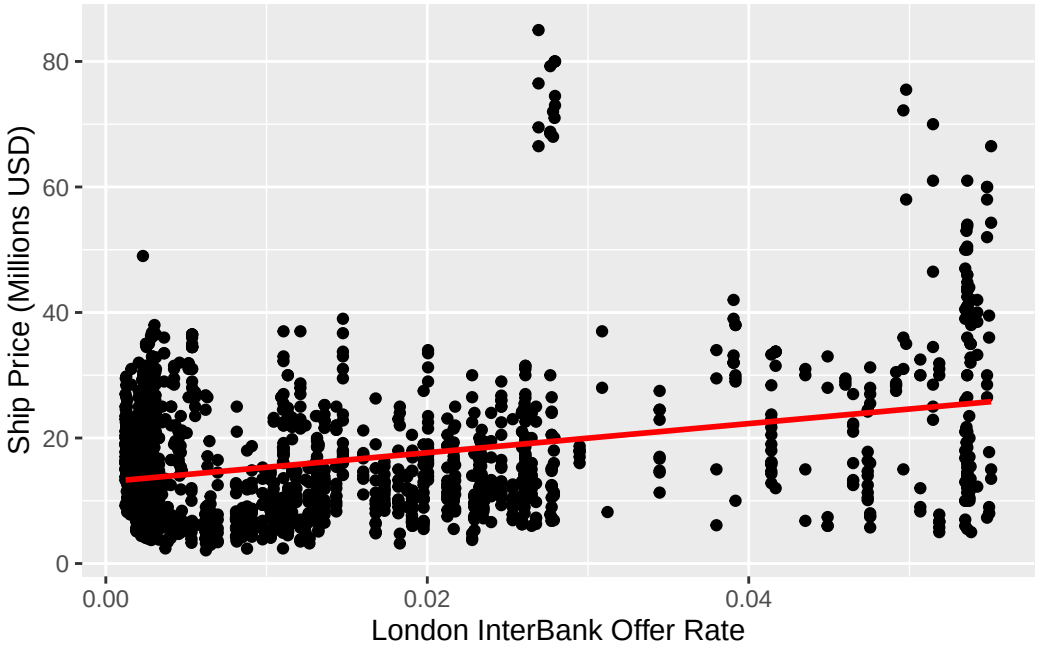



Table 3: Correlation of Ship Price and LIBOR

| pearson | spearman | kendall |
|---|---|---|
| 0.32 | 0.12 | 0.09 |

Figure 4 and Table 3 show a weak positive linear correlation between price and average interest rates over the last 3 months. As with any good, we expect its price to increase with the interest rate, which represents the price to borrow capital. It's clear from Figure 4 that ship prices also vary independently of the `LIBOR`, which is plausible given variations in ship sizes, age, and capacity.

## References

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